

Challenge (Habanero)

- Q1.** A coder uses 18 minutes of data to download a game. If the data usage is 3 minutes per level, how many levels can the coder download with the remaining data after playing 2 levels?
(A) 4
(B) 5
(C) 6
(D) 7
(E) 8
- Q2.** A high-score table can hold 150 entries. If each entry requires 5 bytes of data, how many kilobytes of data can be stored for the high scores?
(A) 0.75
(B) 0.85
(C) 0.95
(D) 1.05
(E) 1.15
- Q3.** To level up, a gamer must collect 48 points. If they have 27 points, how many more points are needed to reach the next level, using 3-point increments?
(A) 7
(B) 8
(C) 9
(D) 10
(E) 11
- Q4.** If 24 players are divided into teams of 4, how many teams can be formed?
(A) 4
(B) 5
(C) 6
- (D) 7
(E) 8
- Q5.** A game uses 15-second intervals for its timer. How many intervals are there in 9 minutes?
(A) 36
(B) 40
(C) 45
(D) 50
(E) 60
- Q6.** If a gamer's score increases by 120 points in 4 games, what is the average score increase per game?
(A) 20
(B) 25
(C) 28
(D) 30
(E) 35
- Q7.** A programmer can write 35 lines of code in 7 minutes. How many minutes does it take to write 5 lines of code?
(A) 0.8
(B) 1
(C) 1.2
(D) 1.4
(E) 1.6
- Q8.** If 99 gamers play 3 rounds, how many total rounds are played using the formula $R = 3G$?
(A) 297
(B) 300
(C) 303
(D) 306
(E) 309

Open Ended Questions (FRQ)

- Q9.** A coding challenge has 12 problems. If each problem is solved in 4 parts, and 3 problems have been solved so far, how many parts remain, showing all division steps and remainders.
- Q10.** A gamer downloads 24 megabytes of data for 4 games, with 8 megabytes left. If 4 friends join, how many more megabytes are needed to have 2 megabytes per player, explaining all calculations with division and remainders.

Chef's Tips

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- Tip 1: Make sure students apply divisibility rules for 2, 3, 5, 9, and 10 correctly.
- Tip 2: Encourage precise long division with remainders for word problems.
- Tip 3: Check for accurate representations of division results in the free response questions.